UNIVERSITY OF MAKATI

J.P Rizal Extension, West Rembo Makati City

COLLEGE OF COMPUTING AND INFORMATION SCIENCES

PROJECT CHARTER

Group Number: 2	**Date:** November 9, 2026	**Revision Number:** 6
Project Name: TakoSync: A Sales-Integrated Inventory Monitoring and Auditing System		
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I. Project Summary and Goals

TakoSync is a sales-integrated inventory monitoring and auditing system designed for Gelo's Takoyaki, a business operating across three branches with two staff members assigned to each branch. The system will consist of an Android application used by both staff and the owner, and a web-based portal accessible only to the owner. The Android application will serve as the point-of-sale (POS) system for recording customer orders and will provide authorized access to sales and inventory information. The web portal will provide the owner with centralized access to business reports, branch comparisons, inventory records, and business configuration settings.

The system addresses common challenges in monitoring ingredient quantities, recording sales during busy hours, and identifying differences between recorded and actual inventory. When staff complete a sale through the Android application, TakoSync will automatically calculate the expected ingredient consumption based on the configured recipe for each product. This will reduce the need for staff to manually deduct ingredients after every transaction. The system will also support stock-in and stock-out recording, waste documentation, low-stock and expiration alerts, and physical inventory reconciliation to identify discrepancies.

The purpose of TakoSync is to improve the consistency and accessibility of sales and inventory information while reducing repetitive manual work. Its main functions include sales transaction recording, automatic ingredient-consumption calculation, inventory monitoring, inventory auditing, and report generation. Both staff and the owner will be able to use the Android application, but their permissions will differ according to their roles. Staff will be limited to the functions and records of their assigned branch, while the owner will have access to authorized information across all branches. The owner-only web portal will provide a more comprehensive view of the business through centralized dashboards, reports, branch comparisons, and configurable system settings. TakoSync falls under the category of business information systems, specifically sales-integrated inventory monitoring and auditing systems.

TakoSync presents an opportunity to improve inventory visibility and support better operational decisions for Gelo's Takoyaki. By connecting sales records with expected ingredient consumption and inventory reports, the system can help the owner identify potential shortages, review recurring discrepancies, and make more informed replenishment decisions. Although the initial implementation will focus on Gelo's Takoyaki, its configurable products, recipes, measurement units, stock thresholds, and business settings may allow it to be adapted to other food kiosks or similar businesses with comparable sales and inventory operations.

II. Project Scope (System Features)

The scope of the project includes the design, development, and testing of TakoSync: A Sales-Integrated Inventory Monitoring and Auditing System for Gelo's Takoyaki. The system will support inventory monitoring and stock auditing across the business's three branches through an Android mobile application and a web management portal. The mobile application will provide

both Staff and Owner access, while the website will be exclusively for the Owner. Both platforms will use a shared database to keep inventory records synchronized.

System Components

A. Owner's Module (Web Management Portal)

User Requirement: The owner shall be able to access the web portal to monitor and compare the performance of all three branches, generate sales and inventory reports, configure products, recipes, batter preparation, manage branch staff accounts and branch assignments, review inventory movements and audit history, and validate physical counts and authorize inventory adjustments after reviewing discrepancies. The portal shall also provide centralized business and inventory configuration to support the business's operations.

1. **Owner Login and Account Access** – The system shall allow only the owner, using authorized credentials, to access the web portal. Staff accounts are prevented from accessing it, and a secure owner session is created after successful authentication.
2. **Centralized Owner Dashboard** – The system shall present total sales, branch sales, inventory status, low-stock and expiration alerts, recent sales and inventory activities, and inventory discrepancies requiring review from all three branches.
3. **Multi-Branch Sales Monitoring** – The system shall allow the owner to view combined sales totals, sales per branch, product sales summaries, and transaction details across all three branches. The owner shall also be able to filter sales information by date or reporting period.
4. **Multi-Branch Inventory Monitoring** – The system shall allow the owner to view branch-specific inventory information, including ingredient and supply quantities, units of measurement, stock status, prepared batter inventory, and expiration information. The system shall also provide a branch comparison view to help the owner monitor inventory across all three branches.
5. **Sales Report Generation** – The system shall allow the owner to generate sales reports for selected reporting periods. Sample reports that can be generated:
    a. **Total Sales Report** – Displays the combined sales of all three branches.
    b. **Branch Sales Report** – Displays the sales of each branch.
    c. **Products Sold Report** – Displays the products sold and their quantities.
    d. **Transaction Summary Report** – Displays transaction summaries and applicable payment information.
6. **Inventory Report Generation** – The system shall allow the owner to generate inventory reports for selected branches and reporting periods. Inventory movement types shall remain separate for auditing and verification. The available reports shall include:
    a. **Expected Stock Quantity Report** – Displays the expected quantity of each ingredient, supply, and prepared batter.
    b. **Stock-In Report** – Displays the recorded received and removed inventory.
    c. **Sales Consumption Report** – Displays the ingredient consumption calculated from completed sales.
    d. **Waste and Spillage Report** – Displays the recorded waste, spillage, and spoilage.
    e. **Physical Inventory Count and Discrepancy Report** – Displays the physical counts and the discrepancies found, including count status and revisions.
    f. **Authorized Adjustment Report** – Displays the adjustments authorized by the owner.
7. **Product and Menu Configuration** – The system shall allow the owner to add and edit products, configure selling prices, availability, and descriptions, and manage the menu items displayed to staff. Product changes do not erase historical sales records.

8. **Ingredient and Recipe Configuration** – The system shall allow the owner to add and edit ingredients and supplies, assign categories, define units of measurement, and configure product recipes and required quantities. Recipe changes apply only to future sales calculations. The batter preparation recipe is configured separately from the prepared batter inventory item. The takoyaki product's recipe uses the prepared batter as its inventory item, not the raw batter ingredients.
9. **Prepared Batter Configuration** – The system shall allow the owner to configure the raw ingredient requirements of one batter batch, the five bottles it produces, and the configured yield per bottle used for calculating batter consumption (25 takoyaki pieces per bottle). Raw ingredients are deducted when a batch is prepared, and completed sales consume prepared batter without deducting them again (a two-stage process: batter preparation, then completed takoyaki sale). The system does not claim to measure the exact batter dispensed from a bottle.
10. **Stock Threshold and Expiration Settings** – The system shall allow the owner to configure minimum stock levels, expiration alert periods, and applicable inventory settings used to generate alerts. Alerts do not automatically generate or place purchase orders.
11. **Inventory Reconciliation and Authorized Adjustments** – The system shall allow the owner to review submitted discrepancies and count correction requests before any adjustment is applied. The owner validates a count by authorizing the adjustment, and the validated actual quantity then overwrites the system's expected quantity. Each adjustment preserves the previous expected quantity, actual quantity, adjusted quantity, reason, approving owner, date and time, and related discrepancy record. Approved count corrections retain the original value, revised value, editor, date, and time, and a count whose adjustment has been authorized can no longer be modified.
12. **Employee Account and Branch Assignment Management** – The system shall allow the owner to manage the six employee accounts (two per branch): create an account using the employee's name as the username and the employee ID as the password, update its information, activate or deactivate it, assign it to its branch, and maintain the branch access restriction. Each employee has one individual account.
13. **Business and Inventory Configuration** – The system shall allow the owner to configure supported business settings, including branch, product, and ingredient information, units of measurement, recipes, batter batch settings, stock thresholds, expiration alert settings, and other supported inventory options.
14. **Inventory Audit History** – The system shall maintain an accessible history of completed sales, Sales Consumption, Stock-In, Waste, Spillage, physical inventory counts (with their revisions), discrepancies, and authorized adjustments, with references, branch information, quantities, dates, and times. Because each employee has an individual account, the history identifies the employee account that recorded each transaction.
15. **Owner Profile, Settings, and Logout** – The system shall allow the owner to view permitted profile information, update permitted account settings, manage available security settings, and securely log out. Logging out ends the active session and returns the owner to the web login page.

B. Staff Module (Mobile Application)

User Requirement: The staff shall be able to log in using their individual employee accounts, access the POS assigned to their branch, view the menu and product information, record customer orders and completed sales, generate receipts, and check their branch's inventory. Staff shall also be able to record batter preparation, received supplies, waste, and spillage; monitor stock levels and expiration dates; conduct physical inventory counts and submit discrepancies for owner review; generate inventory reports; and view their branch's daily sales summary and transaction history. The system shall automatically calculate

ingredient consumption based on configured recipes and successfully recorded sales, update expected inventory quantities, and maintain inventory records throughout daily operations.

1. **Login and Branch Account Access –** The system shall allow each staff member to log in using their individual employee account, with their employee name as the username and employee ID as the initial password. The system shall validate the credentials, verify the employee's assigned branch, and restrict access to that branch. Upon successful login, the POS interface shall be displayed as the main screen, and an authenticated session shall begin for the current business day. Staff may close and reopen the application during the same business day without repeatedly entering their credentials while the session remains valid. The session shall automatically expire at the end of the defined business day, requiring staff to log in again when the next business day begins.
2. **Point-of-Sale (POS) and Sales Recording –** The system shall display a kiosk-style POS interface with selectable product buttons. Staff shall be able to add products with a single tap, adjust quantities, remove items, review the order summary and total, select the payment method, confirm the transaction, and receive a unique transaction reference. Selecting a product only adds it to the current order. Only confirmed and recorded sales trigger ingredient consumption, while cancelled or uncompleted orders do not.
    a. **View Menu and Product Information** – Within the POS, staff shall view owner-configured product names, selling prices, and availability. Unavailable products shall be identified and cannot be added to an order.
    b. **View Branch Inventory** – Within the POS application, staff shall be able to open the inventory view for their assigned branch without losing an active order. The view shall show item name, category, unit, expected quantity, stock status, expiration information, and prepared batter inventory.
    c. **Generate Receipt** – After a sale is successfully recorded, staff shall be able to generate and display a receipt containing the transaction reference, date and time, branch, ordered items, quantities, prices, total, applicable payment details and the employee who processed the sale, so that staff can show, share, or print it for the customer. A receipt shall not be generated as a completed-sale receipt for a cancelled or unrecorded order.
    d. **Generate Daily Sales** – Staff shall be able to view or generate the daily sales summary for their assigned branch. This is a branch-level daily view and does not replace the owner's consolidated sales reports.
3. **Prepared Batter Batch Recording** – The system shall allow staff to record the preparation of a batter batch. The system deducts the configured raw ingredients, creates a prepared batter batch record, and records five prepared batter bottles for the batch. Sales then consume from the prepared batter inventory, so the raw ingredients are not deducted again. This is handled in two separate stages: (1) Batter preparation – the system deducts the configured raw ingredients and records the resulting five bottles of prepared batter; (2) Completed takoyaki sale – the system deducts the expected prepared batter quantity based on the configured yield of 25 pieces per bottle. The takoyaki product's sales-consumption calculation uses prepared batter as its inventory item and does not deduct the raw ingredients a second time.
4. **Stock-In Recording** – The system shall allow staff to record received ingredients and supplies, including quantity, unit of measurement, date, expiration date when applicable, and optional supplier or reference details, to increase expected inventory.
5. **Waste and Spillage Recording** – Waste and Spillage Recording – The staff shall be able to record known waste, spillage, spoilage, and other operational losses by specifying the affected inventory item, quantity, reason or category, date, and

applicable remarks. Unexplained inventory discrepancies shall be handled through physical inventory counting and discrepancy reporting rather than automatically classified as waste.

6. **Low-Stock and Expiration Alerts** – The system shall notify staff when ingredients reach configured minimum stock levels or when recorded expiration dates require attention. These alerts shall be accessible through the appropriate interface or navigation section.
7. **Override Inventory Count (Physical Count)** – The system shall allow staff to select an item, view its expected quantity, and enter the actual quantity based on physical counting, weighing, or measuring through a dedicated inventory section. The system shows whether the count matches, is lower than, or is higher than the expected quantity, and saves it as a physical inventory record. Staff may edit a count while it is still a draft or before submission. After submission, a count may be changed only through an authorized correction or revision process, and the system retains the original value, revised value, editor, date, and time of each change. Once a count has been reviewed and an adjustment authorized, staff can no longer modify the original count directly. The count does not change the expected inventory by itself. Once the owner validates the count through the authorized adjustment process, the validated actual quantity overwrites the system's expected quantity, because the expected quantity is only an estimate based on recorded transactions.
8. **Inventory Reporting** – The system shall create reports that compare the expected inventory quantities with the actual counted quantities and identify differences for review. A discrepancy record is created whenever the quantities differ and is submitted for owner review. The system does not assume the cause of a difference or adjust inventory automatically. Sample reports that can be generated:
    a. **Total Ingredient Quantity Report** – Displays the total expected quantity available for each ingredient.
    b. **Ingredient Category Summary Report** – Shows the combined quantities of ingredients grouped by category, such as meat, dry goods, and condiments.
    c. **Ingredient Stock Status Report** – Presents the current stock status of each ingredient based on its expected quantity and configured minimum stock level.
    d. **Branch Inventory Summary Report** – Provides an overview of the branch's available ingredients, their quantities, and units of measurement.
    e. **Inventory Discrepancy Report** – Displays the item, expected quantity, actual quantity, difference, branch, date and time, submitting branch account, and staff remarks.
    f. **Physical Inventory Count Report** – Displays the counted items with their expected and actual quantities, count status, revision history, and whether each count matches, is lower than, or is higher than expected.
9. **Inventory Transaction History** – The staff shall be able to view and filter the inventory transaction history of their assigned branch by transaction type and date. The history shall include Stock-In, Sales Consumption, Waste and Spillage, Physical Inventory Counts and their revisions, and Authorized Adjustments. Each record shall display applicable transaction references, inventory items, quantities, dates, and times for verification.
10. **Staff Profile and Logout** – The staff shall be able to view their permitted employee account information and assigned branch. The staff shall also be able to securely log out of the application, which shall terminate the active session and return them to the login screen. The system shall automatically expire the session at the end of the defined business day.

C. Owner's Module (Mobile Application)

User Requirement: The owner shall be able to log in to the mobile application, view inventory summaries across all three branches, monitor ingredient, supply, and prepared batter stock levels, receive low-stock and expiration alerts, review physical inventory counts

and identified discrepancies, and authorize inventory adjustments. The owner shall also be able to access inventory transaction records across all three branches to monitor stock movements and verify recorded inventory activities. The application shall provide the owner with an overview of inventory conditions across branches and support inventory monitoring, reconciliation, and auditing.

1. **Owner Login and Account Access** – The system shall allow the owner to log in using authorized owner credentials and provide owner-level access. Staff accounts do not receive owner-level access.
2. **Owner Dashboard** – The system shall display total recorded sales, branch sales summaries, inventory status, low-stock items, expiration alerts, and recent sales and inventory activities across all three branches.
3. **Sales Summaries** – The system shall allow the owner to review recorded sales for selected dates and branches, including products sold, transaction details, and applicable payment information. Sample reports that can be generated:
    a. **Daily Sales Summary** – Displays the total recorded sales for a selected day.
    b. **Weekly Sales Summary** – Displays the total recorded sales for a selected week.
    c. **Monthly Sales Summary** – Displays the total recorded sales for a selected month.
    d. **Branch Sales Summary** – Displays the sales of each branch for the selected period.
    e. **Products Sold Summary** – Displays the products sold and their quantities for the selected period.
4. **Multi-Branch Inventory Monitoring** – The system shall allow the owner to select a branch and view ingredient quantities, units of measurement, stock status, prepared batter inventory, and expiration information, and to compare inventory between branches.
5. **Low-Stock and Expiration Monitoring** – The system shall display alerts from all three branches, including the affected ingredient, branch, expected quantity, minimum stock level, batch information when applicable, and recorded expiration date. Alerts support the owner's decisions and do not automatically determine purchases.
6. **Review Inventory Discrepancies** – The system shall allow the owner to review the expected quantity, actual quantity, difference, branch, date and time, submitting branch account, and available remarks of each discrepancy, and to validate the count and authorize an appropriate adjustment when necessary. Once the count is validated, the validated actual quantity overwrites the branch's expected quantity. The owner also reviews staff requests to correct a submitted count, and approved corrections keep the original and revised values.
7. **View Sales and Inventory Records** – The system shall allow the owner to review historical records across all three branches, kept separate by type. Sample records that can be viewed:
    a. **Sales Transaction Records** – Displays recorded sales transactions of all three branches.
    b. **Sales Consumption Records** – Displays the ingredient consumption calculated from completed sales.
    c. **Stock-In Records** – Displays recorded Stock-In transactions.
    d. **Waste and Spillage Records** – Displays recorded waste, spillage, spoilage, and other losses.
    e. **Physical Inventory Count Records** – Displays the recorded physical counts, their status, and revision history.
    f. **Authorized Adjustment Records** – Displays the adjustments authorized by the owner.

8. **Owner Profile and Logout** – The system shall allow the owner to view permitted account information and securely log out.

Tools To be Used:

1. Documentation and Project Management

- **Google Docs:** For preparing project documentation, requirements, and the project charter.
- **Google Sheets:** For organizing sample inventory data, preparing test records, and documenting testing results.
- **Notion:** For tracking project tasks, milestones, and team progress.

2. Design Tools

- **Figma:** For creating wireframes, user interface mockups, and prototypes for the Android mobile application and web management portal.
- **Canva:** For preparing project presentations and visual documentation when needed.

3. Development Tools

a. Programming Languages and Frameworks

- **React Native 0.81:** For developing the mobile-first TakoSync application, including the Staff Module and available Owner interfaces.
- **React 19:** For building reusable interface components and interactive application screens.
- **TypeScript 5:** For implementing application logic, inventory calculations, data validation, and typed database interactions.
- **Expo SDK 54:** For managing the development environment and running the application across supported platforms.
- **Expo Web and React Native Web:** For running the same application through a web browser without maintaining a separate web application.

b. Integrated Development Environment (IDE)

- **Visual Studio Code (VS Code):** For writing, editing, debugging, and maintaining the frontend source code, Supabase integration, database-related files, and application configuration.

c. Version Control and Collaboration

- **Git:** For tracking source code changes, managing versions, and supporting collaborative development.
- **GitHub:** For hosting the project repository, maintaining version history, and coordinating frontend and backend development.

d. Database, Authentication, and Backend Services

- **Supabase:** For providing PostgreSQL database services, authentication, Row Level Security (RLS), and database functions that support inventory transactions, branch-based data access, and daily physical stock checks.
- **Supabase JavaScript Client (@supabase/supabase-js):** For connecting the application to Supabase Auth and PostgreSQL-backed application services.
- **PostgreSQL and SQL Migrations:** For defining and maintaining the database schema, branch records, inventory tables, security policies, and database functions.
- **Supabase CLI:** For linking the local backend configuration to a Supabase project and applying database migrations.

- **AsyncStorage (@react-native-async-storage/async-storage):** For storing the application's session information and device-level preferences, including the theme preference.

e. User Interface and Styling

- **React Native Styling:** For implementing layouts, colors, spacing, and reusable interface styles across mobile and web platforms.
- **Poppins via Expo Google Fonts:** For providing consistent typography throughout the application.
- **Expo Assets:** For managing and displaying application images and other visual resources.

4. Testing Tools

- **Android Emulator:** For testing the Android application's functionality, navigation, and interface in a simulated mobile environment.
- **Physical Android Mobile Phones:** For evaluating the application's functionality and usability during inventory activities.
- **Web Browsers:** For testing the web-compatible version of the application and checking its layout and functionality.
- **TypeScript Type Checking (npm run typecheck):** For checking TypeScript errors and improving code reliability.
- **Supabase Dashboard and SQL Editor:** For configuring authentication, managing database settings, applying trusted administrative changes, and inspecting database records and security policies.
- **Git and GitHub:** For reviewing code changes and coordinating integration between frontend and backend development.

III. Project Milestones

Identify the significant project milestones, including invoicing dates to the client.

Milestone 1 – Requirement Analysis		
Number	**Deliverables**	**Target Date**
1	**System Request**	October 30, 2026
2	**User Requirements**	
3	**System Requirements**	
4	**User Story**	November 3, 2026

Milestone 2 – Design		
Number	**Deliverables**	**Target Date**
1	**Diagrams**	
	• **Software Architecture**	November 6, 2026
	• **Use Case**	
	• **DFD**	
	• **ERD and Data Dictionary**	
2	**Figma Design**	
2.1	• **Sticker Sheet**	November 3, 2026
2.2	• **High Fidelity - Owner Module (Web)**	
2.2.1	• **Owner Login and Account Access**	
	- Email and Password Fields	

	- Show/Hide Password	
	- Login Button	
	- Loading Screen after Logging In	
	- Invalid Credential Messages and Owner-Only Access	
	- Specific Error Messages (Username Not Found, Incorrect Password) and No-Connection Message	
2.2.2	• **Centralized Owner Dashboard**	
	- Total Sales and Branch Sales	
	- Inventory Status	
	- Low-Stock and Expiration Alerts	
	- Recent Sales and Inventory Activities	
	- Inventory Discrepancies Requiring Review	
2.2.3	• **Multi-Branch Sales Monitoring**	
	- Combined Sales Totals and Sales per Branch	
	- Product Sales Summaries and Transaction Details	
	- Date and Reporting Period Filters	
2.2.4	• **Multi-Branch Inventory Monitoring**	
	- Branch-Specific Inventory Tables	
	- Ingredient Quantities, Units, and Stock Status	
	- Prepared Batter Inventory and Expiration Information	
	- Branch Comparison	
2.2.5	• **Sales Report Generation**	
	- Total Sales, Branch Sales, Products Sold, and Transaction Summary Reports	
	- Reporting Period Selection	
	- Print-Friendly Reports	
2.2.6	• **Inventory Report Generation**	
	- Expected Stock Quantity Report	
	- Stock-In Report	
	- Sales Consumption Report	
	- Waste and Spillage Report	
	- Physical Inventory Count and Discrepancy Report	
	- Authorized Adjustment Report	
	- Count Status and Revision History in the Count and Discrepancy Report	
2.2.7	• **Product and Menu Configuration**	
	- Add and Edit Products	
	- Selling Prices and Availability	
	- Product Descriptions and Menu Items Displayed to Staff	
2.2.8	• **Ingredient and Recipe Configuration**	
	- Ingredient Records, Categories, and Units of Measurement	
	- Recipe Ingredients and Required Quantities per Product	
	- Takoyaki Recipe Using Prepared Batter as the Inventory Item	
2.2.9	• **Prepared Batter Configuration**	
	- Raw Ingredient Requirements per Batch	

	- Bottles per Batch and Configured Yield per Bottle	
	- Configured Yield of 25 Pieces per Bottle	
2.2.10	• **Stock Threshold and Expiration Settings**	
	- Minimum Stock Levels	
	- Expiration Alert Periods	
2.2.11	• **Inventory Reconciliation and Authorized Adjustments**	
	- Expected and Actual Quantities and Variance Review	
	- Count Correction Requests (Original Value and Revised Value)	
	- Approve or Reject Correction Request	
	- Validate Count (Authorize Adjustment) With Required Reason	
	- Adjustment History	
2.2.12	• **Employee Account and Branch Assignment Management**	
	- Create, Update, Activate, and Deactivate Employee Accounts (Username = Employee Name, Password = Employee ID)	
	- Branch Assignment	
2.2.13	• **Business and Inventory Configuration**	
	- Branch, Product, Ingredient, Recipe, Batter Batch, Threshold, and Expiration Settings	
2.2.14	• **Inventory Audit History**	
	- Record Type, Branch, Date Range, and Reference Filters	
	- Reference, Quantity, Date and Time, and Related Records, Including Count Revisions	
	- Employee Account and Branch That Recorded Each Entry	
2.2.15	• **Owner Profile, Settings, and Logout**	
	- Profile Information, Account Settings, and Security Settings	
	- Secure Logout Confirmation	
2.3	• **High Fidelity - Staff Module (Mobile Application)**	
2.3.1	• **Login and Branch Account Access**	
	- Username and Password Fields	
	- Show/Hide Password	
	- Login Button	
	- Loading Screen While Logging In	
	- Specific Error Messages (Username Not Found, Incorrect Password, Inactive Account, Missing Fields, No Connection)	
	- Login Successful Message and Redirect to the POS Screen	
	- Session Stays Active After Closing and Reopening the Application (Daily Login Session)	
	- Automatic Session Expiry at the End of the Business Day and Login Required Again at the Start of the Next Business Day	
	- Manual Logout Option	
	- Inactivity Lock Screen (When the Timeout Setting Is Finalized)	
2.3.2	• **Point-of-Sale (POS) and Sales Recording**	

	- Product Name and Selling Price	
	- Available/Unavailable Status	
	- Unavailable Item State	
	- Order Summary, Quantity Adjustment, and Total	
	- Payment Method and Details	
	- Review and Confirm Transaction	
	- Sale Completed Confirmation with Transaction Reference	
2.3.2.1	• **View Menu and Product Information**	
	- Menu Items with Product Name, Selling Price, and Availability	
	- Product Details	
	- Unavailable Product State	
2.3.2.2	• **View Branch Inventory**	
	- Ingredient List and Search	
	- Category, Expected Quantity, and Unit of Measurement	
	- Stock Status and Expiration Information	
	- Prepared Batter Inventory	
	- Ingredient Details	
2.3.2.3	• **Generate Receipt**	
	- Receipt With Transaction Reference, Branch, Date and Time	
	- Ordered Items, Quantities, Prices, and Total	
	- Payment Method and Amount, and Serving Employee	
	- Print or Share the Receipt	
2.3.2.4	• **Generate Daily Sales**	
	- Total Sales and Number of Transactions for the Current Business Day	
	- Sales per Payment Method	
	- Products Sold and Quantities	
2.3.3	• **Prepared Batter Batch Recording**	
	- Configured Raw Ingredient Quantities per Batch	
	- Batch Reference and Preparation Date	
	- Five Prepared Batter Bottles Result	
	- Save Confirmation	
2.3.4	• **Stock-In Recording**	
	- Ingredient Selection	
	- Quantity Received and Unit of Measurement	
	- Date and Expiration Date When Applicable	
	- Supplier or Reference Details When Applicable	
	- Save Confirmation	
2.3.5	• **Waste and Spillage Recording**	
	- Ingredient Selection and Quantity	
	- Reason/Category, Date, and Remarks	
	- Save Confirmation	
2.3.6	• **Low-Stock and Expiration Alerts**	
	- Low-Stock and Out-of-Stock Ingredients	
	- Minimum Stock Levels	
	- Near-Expiration Items and Expired Batches	
	- Alert Details	
2.3.7	• **Override Inventory Count**	
	- Item Selection and Expected Quantity	

	- Actual Count Input and Unit of Measurement	
	- Quantity Difference (Matches, Lower, or Higher)	
	- Submission Confirmation	
	- Draft Count and Edit Before Submission	
	- Correction Request for a Submitted Count	
	- Original Value, Revised Value, Editor, Date, and Time	
	- Locked Count After an Adjustment Is Authorized	
2.3.8	• **Inventory Reporting**	
	- Total Ingredient Quantity Report	
	- Ingredient Category Summary Report	
	- Ingredient Stock Status Report	
	- Branch Inventory Summary Report	
	- Inventory Discrepancy Report (Expected and Actual Quantity, Difference, Staff Remarks, and Review Status)	
	- Physical Inventory Count Report	
	- Count Status and Revision History in the Physical Inventory Count Report	
2.3.9	• **Inventory Transaction History**	
	- Stock-In, Sales Consumption, Waste, Spillage, Physical Inventory Count, and Authorized Adjustment Records	
	- Date and Transaction Type Filters	
	- Count Revision History	
2.3.10	• **Staff Profile and Logout**	
	- Employee Account Information and Assigned Branch	
	- Secure Logout Confirmation	
	- Logout Ends the Daily Session	
2.4	• **High Fidelity - Owner Module (Mobile Application)**	
2.4.1	• **Owner Login and Account Access**	
	- Email and Password Fields	
	- Show/Hide Password	
	- Login Button	
	- Loading Screen after Logging In	
	- Invalid Credential Messages and Access Error States	
	- Specific Error Messages (Username Not Found, Incorrect Password) and No-Connection Message	
2.4.2	• **Owner Dashboard**	
	- Total Recorded Sales and Branch Sales Summaries	
	- Inventory Status	
	- Low-Stock Items and Expiration Alerts	
2.4.3	• **Sales Summaries**	
	- Daily, Weekly, and Monthly Sales Summaries	
	- Branch Sales Summary	
	- Products Sold Summary	
	- Transaction Details and Payment Information	
2.4.4	• **Multi-Branch Inventory Monitoring**	
	- Branch Selection	
	- Ingredient Quantities and Units of Measurement	

	- Stock Status, Prepared Batter Inventory, and Expiration Information	
	- Branch Comparison	
2.4.5	• **Low-Stock and Expiration Monitoring**	
	- Affected Ingredient and Branch	
	- Expected Quantity and Minimum Stock Level	
	- Batch Information and Recorded Expiration Date	
2.4.6	• **Review Inventory Discrepancies**	
	- Expected Quantity, Actual Quantity, and Difference	
	- Branch, Date and Time, and Submitting Employee Account	
	- Staff Remarks and Review Status	
	- Validate Count (Authorize Adjustment) With Required Reason	
	- Review Count Correction Requests (Original Value and Revised Value)	
	- Approve or Reject Correction Request	
2.4.7	• **View Sales and Inventory Records**	
	- Sales Transaction and Sales Consumption Records	
	- Stock-In Records	
	- Waste and Spillage Records	
	- Physical Inventory Count and Authorized Adjustment Records	
	- Count Status and Revision History	
2.4.8	• **Owner Profile and Logout**	
	- Owner Information and Account Details	
	- Secure Logout Confirmation	

Milestone 3 – Development		
Number	**Deliverables**	**Target Date**
1	**Functional Components**	
	Owner Module (Web Portal)	
1.1	• **Owner Login and Account Access**	
	- Validate Owner Credentials and Restrict Access to the Owner Only	
	- Create Secure Owner Sessions	
	- Return Specific Error Messages for an Unregistered Username and an Incorrect Password	
1.2	• **Centralized Owner Dashboard**	
	- Retrieve Consolidated Sales and Inventory Summaries, Branch Status, and Alerts	
	- Retrieve Recent Activities and Discrepancies Requiring Review	
1.3	• **Multi-Branch Sales Monitoring**	
	- Retrieve Combined Sales Totals, Sales per Branch, and Product Sales Summaries	
	- Filter by Date and Reporting Period	
1.4	• **Multi-Branch Inventory Monitoring**	
	- Retrieve Branch-Specific Inventory, Prepared Batter Inventory, and Expiration Information	

	- Compare Branches	
1.5	• **Sales Report Generation**	
	- Generate Sales Reports for Selected Reporting Periods	
	- Generate Total Sales, Branch Sales, Products Sold, and Transaction Summary Reports	
1.6	• **Inventory Report Generation**	
	- Generate the Expected Stock, Stock-In, Sales Consumption, Waste and Spillage, Count and Discrepancy, and Adjustment Reports	
	- Include Count Status and Revision History in the Count and Discrepancy Report	
1.7	• **Product and Menu Configuration**	
	- Add and Edit Products, Prices, Availability, and Descriptions	
1.8	• **Ingredient and Recipe Configuration**	
	- Add and Edit Ingredients, Categories, and Units of Measurement	
	- Configure Recipe Quantities	
	- Configure the Takoyaki Recipe to Use Prepared Batter as Its Inventory Item	
1.9	• **Prepared Batter Configuration**	
	- Configure Raw Ingredient Requirements, Bottles per Batch, and Yield per Bottle	
	- Set the Yield to 25 Pieces per Bottle	
1.10	• **Stock Threshold and Expiration Settings**	
	- Configure Minimum Stock Levels and Expiration Alert Periods	
1.11	• **Inventory Reconciliation and Authorized Adjustments**	
	- Review Discrepancies Before Applying an Adjustment	
	- Review, Approve, or Reject Count Correction Requests	
	- Validate the Count by Authorizing an Adjustment With a Required Reason	
1.12	• **Employee Account and Branch Assignment Management**	
	- Create, Update, Activate, and Deactivate Employee Accounts (Username = Employee Name, Password = Employee ID)	
	- Assign Each Account to Its Branch	
1.13	• **Business and Inventory Configuration**	
	- Configure Branch, Product, Ingredient, Recipe, Batter Batch, Threshold, and Expiration Settings	
1.14	• **Inventory Audit History**	
	- Retrieve and Filter the History of Sales, Stock Movements, Counts, Discrepancies, and Authorized Adjustments	
	- Identify the Employee Account, Branch, Reference, Quantity, Date, and Time	
	- Include Count Revisions in the History	
1.15	• **Owner Profile, Settings, and Logout**	
	- Retrieve Profile Information and Update Permitted Account and Security Settings	
	- Securely End the Session	

2	**Staff Module (Mobile Application)**	
2.1	• **Login and Branch Account Access**	
	- Validate the Username (Employee Name) and Password (Employee ID) Against the Employee Accounts	
	- Display a Loading Screen While the Login Is Processed and Prevent Duplicate Requests	
	- Return Specific Error Messages for an Unregistered Username, an Incorrect Password, an Inactive Account, Missing Fields, and No Connection	
	- Create Secure Sessions	
	- Retrieve the Assigned Branch and Restrict Access to Other Branches	
	- Display the Login Successful Message and Redirect Authenticated Staff to the POS Screen	
	- Keep the Session Active for the Business Day and Restore It After the Application Is Closed and Reopened	
	- Automatically Expire the Session at the End of the Defined Business Day and Require Login Again When the Next Business Day Begins	
	- Allow Staff to Log Out Manually	
	- Apply an Inactivity Lock as an Additional Security Measure (When the Timeout Setting Is Finalized)	
2.2	• **POS and Sales Recording**	
	- Display Menu Items and Add, Adjust, and Remove Order Quantities	
	- Calculate Order Totals and Validate Order Details	
	- Record Payment Details	
	- Save Confirmed Sales with a Unique Transaction Reference	
2.2.1	• **View Menu and Product Information**	
	- Retrieve Product Names, Selling Prices, and Availability Status	
	- Prevent Unavailable Products from Being Selected for a Sale	
2.2.2	• **View Branch Inventory**	
	- Retrieve Ingredient Names, Categories, Expected Quantities, Units, and Stock Status for the Assigned Branch	
	- Retrieve Expiration Information and Prepared Batter Inventory	
2.2.3	• **Generate Receipt**	
	- Retrieve the Confirmed Sale and Build the Receipt (Reference, Branch, Date and Time, Items, Total, Payment, Serving Employee)	
	- Display, Print, or Share the Receipt	
2.2.4	• **Generate Daily Sales**	
	- Compute the Total Sales and Number of Transactions of the Assigned Branch for the Current Business Day	
	- Retrieve the Sales per Payment Method and the Products Sold With Quantities	
2.3	• **Prepared Batter Batch Recording**	

	- Retrieve the Configured Raw Ingredient Requirements for One Batch	
	- Save the Prepared Batter Batch Record	
	- Record Five Prepared Batter Bottles per Batch	
2.4	• **Stock-In Recording**	
	- Validate Ingredient and Quantity Inputs	
	- Record Received Supplies, Expiration Dates, and Supplier or Reference Details	
	- Increase Expected Inventory and Preserve the Transaction Record	
2.5	• **Waste and Spillage Recording**	
	- Record the Ingredient, Quantity, Reason or Category, Date, and Remarks	
	- Reduce Expected Inventory and Preserve the Transaction Record	
2.6	• **Low-Stock and Expiration Alerts**	
	- Compare Expected Inventory Quantities Against Configured Minimum Stock Levels	
	- Identify Low-Stock and Out-of-Stock Ingredients	
	- Retrieve Expiration Dates and Identify Near-Expiration Ingredients and Expired Batches	
2.7	• **Override Inventory Count**	
	- Retrieve the Expected Quantity of the Selected Item	
	- Record the Actual Count and Compute the Difference	
	- Identify Whether the Count Matches, Is Lower, or Is Higher Than Expected	
	- Save the Physical Inventory Record	
	- Save the Count as a Draft and Allow Editing Before Submission	
	- Allow Changes After Submission Only Through a Correction Request	
	- Show the Revision History (Original Value, Revised Value, Editor, Date, and Time)	
	- Prevent Direct Modification of a Count Once Its Adjustment Is Authorized	
2.8	• **Inventory Reporting**	
	- Display Quantity Differences and Remarks for Owner Review	
	- Submit Discrepancies With Staff Remarks to the Owner	
	- Generate the Sample Inventory Reports of the Staff Module	
2.9	• **Inventory Transaction History**	
	- Retrieve Stock-In, Sales Consumption, Waste, Spillage, Physical Inventory Count, and Authorized Adjustment Records	
	- Filter Records by Transaction Type and Date	
	- Show the Revision History of Physical Inventory Counts	
2.10	• **Staff Profile and Logout**	
	- Retrieve Employee Account Information and the Assigned Branch	
	- Securely End the Session	

	- End the Daily Session on Manual Logout and Require the Employee Credentials Again	
3	**Owner Module (Mobile Application)**	
3.1	● **Owner Login and Account Access**	
	- Validate Owner Credentials and Verify the Owner Role	
	- Display a Loading Screen and Return Specific Error Messages for an Unregistered Email and an Incorrect Password	
	- Manage Secure Sessions	
3.2	● **Owner Dashboard**	
	- Retrieve Consolidated Sales and Inventory Summaries Across All Three Branches	
	- Retrieve Low-Stock and Expiration Alerts and Recent Activities	
3.3	● **Sales Summaries**	
	- Retrieve Daily, Weekly, and Monthly Sales Totals	
	- Retrieve Branch Sales, Products Sold, Transaction Details, and Payment Information	
3.4	● **Multi-Branch Inventory Monitoring**	
	- Retrieve Expected Inventory Quantities, Units, and Stock Status for Each Branch	
	- Retrieve Prepared Batter Inventory and Expiration Information	
	- Compare Inventory Between Branches	
3.5	● **Low-Stock and Expiration Monitoring**	
	- Identify Ingredients Below Their Configured Minimum Stock Levels	
	- Retrieve and Display Near-Expiration and Expired Items with Branch and Batch Information	
3.6	● **Review Inventory Discrepancies**	
	- Retrieve Expected Quantities, Actual Counts, Differences, Staff Remarks, and Review Status	
	- Authorize Adjustments When Necessary	
	- Review, Approve, or Reject Count Correction Requests	
3.7	● **View Sales and Inventory Records**	
	- Retrieve Sales Transactions and Sales Consumption Records	
	- Retrieve Stock-In Records, Waste and Spillage, Physical Counts, and Authorized Adjustments	
	- Retrieve Count Status and Revision History	
3.8	● **Owner Profile and Logout**	
	- Retrieve Owner Account Information	
	- Securely End the Session	
4	**Backend Features (Supporting System Processes)**	
4.1	● **Automatic Ingredient Consumption Calculation**	
	- Calculate and Deduct Expected Ingredient Usage From Confirmed Sales Using the Configured Recipes	
	- Record Sales Consumption Under the Sale's Unique Transaction Reference and Prevent Duplicate Deductions	
	- Flag Failed Inventory Updates for Retry or Review While Keeping the Sale Recorded	

4.2	• **Prepared Batter Two-Stage Deduction**	
	- Deduct Raw Ingredients and Add Five Prepared Batter Bottles at Batter Preparation	
	- Deduct Prepared Batter (Pieces Sold / 25-Piece Yield per Bottle) at Takoyaki Sale Without Deducting Raw Ingredients Again	
4.3	• **Session and Access Security**	
	- Daily Login Session Handling, Expiry, and Inactivity Lock	
	- Branch-Level Access Restriction Using Row Level Security	
4.4	• **Count, Discrepancy, and Adjustment Processing**	
	- Create Discrepancy Records, Keep Count Revisions, and Lock Validated Counts	
	- Overwrite the Expected Quantity With the Validated Actual Quantity and Preserve the Adjustment Details	
4.5	• **Audit Trail, Alerts, and Data Integrity**	
	- Maintain the Audit Trail and Keep Transaction Types Separate	
	- Generate Low-Stock and Expiration Alerts and Synchronize Records Across Mobile and Web	

Milestone 4 – Integration and Testing		
Number	**Deliverables**	**Target Date**
1	**Functional Component Testing**	
1.1	• **Test Owner Web Portal Features**	
	- Owner Login and Account Access (Specific Error Messages and Loading Screen)	
	- Centralized Owner Dashboard	
	- Multi-Branch Sales and Inventory Monitoring	
	- Sales and Inventory Report Generation	
	- Product, Menu, Ingredient, and Recipe Configuration	
	- Prepared Batter Configuration	
	- Stock Threshold and Expiration Settings	
	- Inventory Reconciliation and Authorized Adjustments	
	- Employee Account and Branch Assignment Management	
	- Business and Inventory Configuration	
	- Inventory Audit History	
	- Owner Profile, Settings, and Logout	
1.2	• **Test Staff Module Features**	
	- Login and Branch Account Access (Specific Error Messages, Loading Screen, Daily Login Session Persistence, Expiry, Re-login, Manual Logout, and Inactivity Lock)	
	- POS and Sales Recording	
	- View Menu and Product Information	
	- View Branch Inventory	
	- Generate Receipt	
	- Generate Daily Sales	

	- Prepared Batter Batch Recording	
	- Stock-In Recording	
	- Waste and Spillage Recording	
	- Low-Stock and Expiration Alerts	
	- Override Inventory Count	
	- Count Draft, Editing, Correction Process, and Locking After Authorization	
	- Inventory Reporting	
	- Inventory Transaction History	
	- Staff Profile and Logout	
1.3	• **Test Owner Mobile Features**	
	- Owner Login and Account Access (Specific Error Messages and Loading Screen)	
	- Owner Dashboard	
	- Sales Summaries	
	- Multi-Branch Inventory Monitoring	
	- Low-Stock and Expiration Monitoring	
	- Review Inventory Discrepancies	
	- Count Validation and Correction Request Review	
	- View Sales and Inventory Records	
	- Owner Profile and Logout	
1.4	• **Test Backend Features**	
	- Automatic Ingredient Consumption Calculation	
	- Prepared Batter Two-Stage Deduction	
	- Session and Access Security	
	- Count, Discrepancy, and Adjustment Processing	
	- Audit Trail, Alerts, and Data Integrity	
2	**Integration Testing**	
	- Test POS-to-Inventory Integration	
	- Test Sales-Based Ingredient Consumption	
	- Test Receipt and Daily Sales Against Recorded Sales	
	- Test Prepared Batter Preparation and Consumption	
	- Test Two-Stage Batter Deduction (Raw Ingredients at Preparation, Prepared Batter at Sale) Without Double Deduction	
	- Test Duplicate Deduction Prevention and Failed Inventory Update Handling	
	- Test Stock-In Integration	
	- Test Physical Count and Discrepancy Workflow	
	- Test Count Editing, Correction, and Validation That Overwrites the Expected Inventory	
	- Test Owner Adjustment and Audit History Integration	
	- Test Mobile and Web Synchronization	
	- Test Daily Session Persistence and Expiry Across App Restarts and the Business Day Cut-Off	
3	**Non-Functional Testing**	
	- Usability Testing	
	- Role-Based Access Control Testing	
	- Authentication and Session Security Testing	
	- Compatibility Testing	
	- System Reliability and Validation Testing	

Milestone 5– Deployment and Finalization		
Number	**Deliverables**	**Target Date**
1	**System Hosting and Setup**	
2	**Final Project Documentation**	
3	**System demonstration and turn over**	

IV. Project Organizational Structure

No.	Name	Role	Responsibilities	Contact Details
1	Denise Mae Lucero	Project Manager – Researcher	1.Oversee the overall planning, progress, and completion of the project. 2.Assign tasks and ensure that all team members meet their responsibilities and deadlines. 3.Monitor the project's progress and help resolve issues that may affect development. 4.Gather and analyze information about user needs, similar applications, and existing systems. 5.Provide research insights and suggestions to improve the application's features and overall development.	dlucero.8784@umak.edu.ph
2	Marjea Marbibi	Front-End Developer – Back-End Developer	1.Lead the overall frontend development and implementation of the application. 2.Design, develop, and maintain user-facing features and interactive interfaces based on the approved UI/UX designs. 3.Handle backend development tasks, including database integration, application logic, and system functionalities. 4.Integrate frontend components with backend services to ensure proper system functionality. 5.Oversee the organization, consistency, and overall implementation of the application's development. 6.Collaborate with other team members to ensure that the application meets its functional and technical requirements. 7.Debug, maintain, and improve the application's frontend and backend components.	mmarbibi.k12255990@umak.edu.ph
3	Michelle Ann Sahorda	Back-End Developer – Front-End Development Support	1.Focus primarily on backend development, including database management, backend logic, and system functionalities. 2.Develop, maintain, and fix backend services to ensure proper system operations.	msahorda.8785@umak.edu.ph

			3.Identify, debug, and resolve backend errors and integration issues. 4.Assist in connecting frontend components with backend services and database operations. 5.Provide frontend development assistance when needed to support the overall implementation of the application. 6.Collaborate with the primary developer to ensure smooth integration between frontend and backend components. 7.Help maintain the application's performance, reliability, and data security.	
4	Isabel San Esteban	UI/UX Designer – Front-End Designer	1.Design user-friendly, visually appealing, and consistent layouts for the application. 2.Create wireframes, mockups, prototypes, and other visual designs for the system. 3.Focus on providing a simple, smooth, and easy-to-use experience for users. 4.Design and implement frontend elements based on the approved UI/UX design. 5.Work closely with the developers to ensure that designs are properly implemented. 6.Suggest improvements to enhance usability, accessibility, and user engagement.	isanesteban.8783@umak.edu.ph
5	Lorraine Jade Zarate	Quality Assurance – Tester	1.Test the application's features and functionalities to identify bugs and errors. 2.Ensure that the system meets the project's requirements and works as expected. 3.Conduct different types of testing, including functionality, usability, and user experience testing. 4.Document and report identified issues clearly to the development team. 5.Verify that reported bugs and issues are properly fixed before deployment. 6.Suggest improvements to improve the application's	lzarate.7629@umak.edu.ph

			quality, usability, and overall performance.	
6	Prof. Era Marie Gannaban	- SoftEng1 Adviser	The adviser provides technical advice and guidance to improve the software development processes. The responsibilities are as follows: 1. Assess and analyze the clients' existing processes and suggest recommendations to optimize the software development workflows. 2. Assist in defining project goals, milestones, and timelines. 3. Advise on the selection of appropriate programming languages, frameworks and tools for designing and development. 4. Establish quality assurance processes to catch and fix issues early in the development life cycle.	
7	Prof. Era Marie Gannaban	- HCI Adviser	Provides technical advice and guidance for the following: 1. Identify usability issues and areas for enhancement. 2. Understand the needs and preferences of the target users. 3. Helps identify the personas and user flows. 4. Provides recommendations to improve usability and user engagement.	